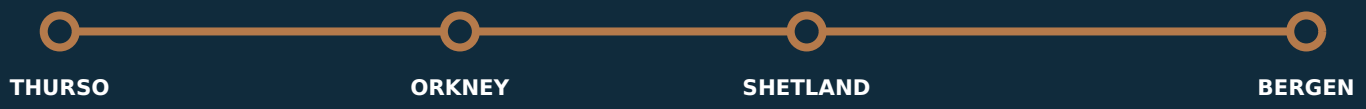
PROPOSED UK-NORWAY PUBLIC-PRIVATE PARTNERSHIP

NORTH SEA HYDROGEN LINK

Thurso / Orkney / Shetland / Bergen

Development prospectus

Sponsor discussion draft | Version 1.0 | 5 November 2026



Strategic corridor diagram only. No surveyed alignment or pipeline length.

A dedicated hydrogen transport corridor between northern Scotland and western Norway, with private operating responsibility, public-interest governance and investment tied to contracted demand.

DOCUMENT STATUS

This is a concept-stage project prospectus, not an approved securities prospectus, offer of securities, financing commitment or invitation to subscribe. No project company, government partner, developer, buyer or lender is represented as appointed or committed. The next decision is whether to commission a bounded feasibility programme.

INDEPENDENT CONCEPT / NOT A BOOKABLE OR COMMITTED INFRASTRUCTURE PROJECT

01 / INVESTMENT PROPOSITION

A corridor worth testing

Develop the evidence first. Commit construction capital only when the route, customers and financing support one another.

The North Sea Hydrogen Link would transport gaseous hydrogen along a broad Thurso-Orkney-Shetland-Bergen corridor. It would use an independently engineered pipeline, with landfalls and potential island connections selected by surveys and demand. Following the rail corridor is a strategic starting point, not a requirement to share railway tunnels or structures.

Proposed term	Development position
Project vehicle	North Sea Hydrogen Link SPV: working name; entity and jurisdiction not yet established.
Sponsor equity	60% competitively selected energy consortium; 40% public equity. Illustrative public allocation: 20% UK and 20% Norway, subject to lawful authority and agreement.
Scope	Dedicated pipeline, necessary reception/compression, metering, controls, landfalls and integrity systems. Production, storage and onward networks are separately contracted interfaces.
Operating term	Proposed 30-year term after commissioning; legal form, asset ownership and end-of-term obligations to be negotiated.
Revenue	Long-term capacity reservations plus transport charges; no assumed commodity trading margin or taxpayer guarantee.
Immediate decision	Explore sponsor participation and approve a capped feasibility mandate; no construction or fundraising launch authorised by this document.

What is established - and what is proposed

UK policy recognises hydrogen transport as part of infrastructure planning, while DESNZ retains decisions on business-model support [1]. This does not confer support on this route. Norway has commissioned other international hydrogen value-chain studies [3]; those studies do not establish this corridor's feasibility.

No route survey, customer book, capacity, cost estimate, tariff, return forecast, delivery date or approved financing package exists within this proposal. Ownership percentages are negotiating assumptions, not commitments.

02 / ROUTE AND SCOPE

Follow the corridor, test alternatives

A pipeline must stand on its own business case. Neither the speculative railway nor a shared construction programme is assumed to be available.

Section	Work required before route selection
Thurso landfall	Compare reception and injection options, electricity supply, access, nearby users and any onward mainland pipeline. A Thurso terminal alone does not provide UK-wide hydrogen distribution.
Orkney interface	Assess whether a landfall and island connection justify added construction and operating cost. Local production and demand must be evidenced; no terminal is mandatory.
Shetland interface	Compare a hub, branch connection or offshore bypass. Secure land, power and marine access only if an island interface creates net value.
Bergen region	Select a landfall and terminal near contracted Norwegian supply or demand. The working name does not prescribe central Bergen or guarantee onward network access.
Offshore crossings	Investigate bathymetry, seabed stability, geology, fisheries, shipping, cables, existing assets and protection/repair options. Pipeline and railway alignments may diverge.

Options appraisal

Compare the island-hub corridor with an independently routed offshore pipeline, phased domestic links and alternative energy-delivery options, including electricity transmission or shipped hydrogen derivatives where relevant. Retain the requested corridor only if it passes a whole-system value comparison.

Scope boundary

No hydrogen pipeline is proposed inside passenger or railway rescue bores. Any crossing or shared landfall requires a documented interface agreement and hazard assessment. No shared trench, tunnel, bridge deck, emergency access or rail-dependent maintenance saving is included in the financial illustration.

Distance discipline: do not reuse lengths from the earlier fictional railway timetable as pipeline survey data. This prospectus deliberately leaves total length unquantified.

03 / MARKET AND DEMAND

Contract the use, not just the pipe

The central commercial question is who will pay for dependable capacity, at what volume and for how long.

Potential customers include industrial hydrogen consumers, chemical or fuel producers, hydrogen production projects and storage operators. These are target classes, not identified anchor customers. Electricity supply, hydrogen production and downstream conversion economics must be assessed alongside transport.

Evidence package	Minimum decision-useful content
Supply	Producer location; commissioning dependencies; annual and hourly supply; carbon intensity; purity; pressure; availability; delivered cost and financial standing.
Demand	Named users; process conversion investment; annual and peak needs; willingness to pay; alternative fuels; credit quality and enforceable booking terms.
Network interfaces	Entry and exit points; compatible specifications; onward transmission/storage capacity; balancing and emergency arrangements.
Open season	Transparent market sounding, followed by conditional capacity requests and binding contracts before FID. Avoid counting the same demand at several potential projects.
Climate evidence	Lifecycle emissions and electricity sourcing, including compression and leakage. Transporting hydrogen does not by itself establish low-carbon performance.

Initial commercial position

Prefer a pipeline transport business to speculative merchant hydrogen trading. Determine the initial direction of flow from contracted supply and demand; bidirectionality is an option requiring separate design and market justification. Treat production and storage investment as separate unless explicitly included after appraisal.

Demand gate

Before selecting design capacity, obtain independently reviewed commitments with sufficient tenor, volume and credit support to support the proposed capital structure under downside conditions. The required booking threshold will emerge from lender testing; no unsupported universal percentage is prescribed.

Policy context: near-term UK planning emphasises connecting production, storage and credible industrial/power demand [1]. International value-chain studies also depend on market development and technology qualification [3].

04 / TECHNICAL AND SAFETY BASIS

Independent hydrogen infrastructure

A hydrogen-qualified system, with maintainable offshore assets and independently reviewed separation from the railway.

Design workstream	Required development output
Capacity and hydraulics	Hourly flow envelope; inlet/outlet conditions; compression energy; pressure drop; linepack; transient response and operating modes.
Materials and fabrication	Hydrogen-specific assessment of pipe, welds, fittings, seals and equipment; qualification for fatigue, fracture, corrosion and manufacturing quality.
Route and protection	Geotechnical evidence; installation method; stability and burial/protection design; crossings; landfalls; inspection and credible repair strategy.
Control and isolation	Metering and product-quality controls; leak detection; isolation philosophy; communications resilience; cyber protection and emergency coordination.
Assurance and operations	Independent verification; testing and commissioning; integrity management; inspection intervals; trained operators; spares and decommissioning plan.

Standards and regulatory interfaces

DNV-RP-F123, edition 2026-03, addresses hydrogen pipeline structural integrity and the effect of hydrogen on steel pipe and connecting welds; it is used with referenced pipeline standards and is not a complete component specification [4]. A project design code and qualification programme remain to be selected.

Scottish guidance identifies pipeline safety duties, HSE oversight and additional authorisation considerations for hydrogen pipelines [2]. The applicable offshore, onshore, UK-Norwegian and cross-border regimes need a project-specific legal and regulator engagement plan. This is not a permit determination.

Design freeze is deferred

No diameter, pressure, throughput, compression duty, separation distance or operating envelope is fixed here. Final values depend on hydraulic, materials, safety and route studies. The prospectus does not assert that an extreme subsea railway or pipeline configuration is buildable.

05 / PARTNERSHIP AND GOVERNANCE

Public purpose, private delivery

Shareholder group	Illustrative equity share	Proposed role
Energy consortium	60%	Lead development, technology integration, construction procurement, finance and long-term operation.
UK public investor(s)	20%	Policy coordination and approved equity participation; Scottish and island interests represented through agreed arrangements.
Norwegian public investor(s)	20%	Norwegian coordination and approved equity participation; relevant national and local stakeholders engaged.

Equity is not the funding stack

These percentages allocate subscribed sponsor equity only. They do not imply that governments pay 40% of total construction cost, guarantee project debt or accept unlimited cash calls. Debt, grants, equity and reserves are separate financing decisions; no leverage ratio is assumed.

Proposed governance terms

Use proportionate board representation with independent audit/risk oversight and a qualified safety committee. Reserve decisions on budgets, material debt, asset disposal, related-party contracts, tariff policy, changes to purpose and public liabilities for defined shareholder consent. Keep regulator decisions independent of investor control.

Appoint the lead private developer through a legally compliant competitive process. Publish selection criteria covering hydrogen/offshore competence, balance sheet, operating capability, commercial offer and community outcomes. Manage supplier and shipper conflicts through disclosed related-party rules.

Public exposure and accountability

Each funding stage has a defined contribution cap, reporting obligations and stop-work rights. Any guarantee, minimum revenue or termination compensation must be separately priced, authorised and disclosed. Specify responsibility for cost overruns, abandonment, remediation and decommissioning; caps do not extinguish statutory duties.

The proposed operating term begins at acceptance, not at signing. End-of-term ownership, handback quality, residual value, environmental liabilities and any concession extension require explicit agreement. No government or company commitment is represented.

06 / REVENUE AND FINANCING

A transport business with clear boundaries

Commercial component	Proposed approach
Capacity charge	Long-term ship-or-pay reservation, subject to agreed availability, curtailment, force-majeure and credit provisions. Negotiated terms, not guaranteed revenue.
Usage charge	Recover efficient variable energy and throughput costs through transparent measures. Define quality, imbalance, losses and metering responsibilities.
Access	Provide transparent, non-discriminatory access consistent with the applicable regime. Allocate expansion costs fairly; do not promise unlimited spare capacity.
Credit support	Customer guarantees or security proportionate to risk; replacement rights and lender-reviewable contract duration.
Public support	Only if separately approved, with explicit limits and clawbacks/performance conditions. Do not presume UK regional hydrogen programmes cover this international corridor.

Financing sequence

Fund early development with risk equity and any expressly approved capped public contribution. Seek construction finance only after route, cost, contracts, consents and technical assurance mature. Match currency exposure across GBP/NOK/EUR revenue, construction and debt; avoid an unhedged mismatch disguised as project return.

Cash-flow priorities to negotiate

Operating costs and statutory obligations; debt service and required reserves; lifecycle and decommissioning funding; then permitted distributions. Actual priority, security and insolvency treatment require financing counsel and negotiated documents.

The bankability test

Run construction-delay, capex overrun, low-booking, shipper-default, energy-price, interest-rate and exchange-rate scenarios. Demonstrate debt coverage and liquidity on contracted revenues before assuming equity distributions. The illustration on page 8 is not a lender model or an investment return calculation.

UK hydrogen infrastructure planning and DESNZ support decisions provide context, not an award or regulatory entitlement for this scheme [1].

07 / FINANCIAL SENSITIVITY

What utilisation does to unit cost

Illustrative arithmetic only. None of the capital or capacity values below is an estimate of this pipeline.

To show why contracted demand matters, assume 0.50 million tonnes/year of nominal transport capacity, a 30-year annuity at 8%, and fixed operating/lifecycle costs equal to 2% of initial capital per year. These are analyst-selected test inputs, not proposed financing terms, engineering outputs or a costed base case.

Test capital input	40% utilisation	70% utilisation	90% utilisation
£2bn	£1.09/kg	£0.62/kg	£0.48/kg
£4bn	£2.18/kg	£1.24/kg	£0.97/kg
£6bn	£3.26/kg	£1.87/kg	£1.45/kg

Annual fixed cost recovery per kilogram actually transported. Nominal throughput at 40%, 70% and 90% is 0.20, 0.35 and 0.45 million tonnes/year respectively. Figures exclude variable compression electricity, taxes, working capital, transport losses and any extra decommissioning cost beyond the assumed lifecycle allowance.

Reproducible calculation

Capital recovery factor = $r(1+r)^n / [(1+r)^n - 1] = 0.08883$, using r = 8% and n = 30 years.
Annual fixed revenue requirement = capital input × (capital recovery factor + 2%).
Unit fixed recovery = annual fixed requirement / annual kg transported.

Worked test case

At the £4bn test input and 70% utilisation: annual fixed recovery is approximately £435.3m. Dividing by 350 million kg gives £1.24/kg, before excluded costs. If variable energy were 2 kWh/kg at £60/MWh, it would add £0.12/kg; both energy figures are additional assumptions, not a compressor design or power quote.

Interpretation

Lower utilisation raises the recovery burden per delivered kilogram sharply. Capacity charges can shift some volume risk to contracted shippers, but cannot create demand or eliminate counterparty default. A larger pipe may increase capital and reduce future congestion; this trade-off needs a costed hydraulic and demand model.

No IRR, debt capacity, equity subscription price, net present value or construction budget is claimed. An 8% annuity is a simplifying recovery assumption, not a promised investor return. Figures are nominal-style test arithmetic without inflation; use a consistent real/nominal basis in the eventual model.

08 / RISK ALLOCATION

Name the risk and its owner

Principal risk	Initial allocation / control	Evidence before FID
Insufficient supply or demand	Sponsors develop the market; shippers commit capacity. Public sector does not automatically underwrite utilisation.	Binding volumes, credit support and credible alternatives analysis.
Route / geology / installation	Sponsors investigate; transfer defined risks to capable contractors where priceable. Retain and fund residual risks openly.	Surveys, installation plan, probabilistic cost/schedule risk.
Rail interface / dependency	Pipeline SPV owns interface obligations. No assumed rail delivery or common emergency facilities.	Separation case and independently operable design.
Hydrogen integrity / leakage	Operator and competent suppliers carry contractual duties; statutory responsibilities remain.	Qualification, monitoring, repair capability and assurance.
Construction / delay	Private-led delivery with realistic performance security; uninsurable exposures explicitly allocated.	Procurement package, guarantees, contingencies and reserves.
Consents / cross-border rules	Governments coordinate; independent authorities decide. Sponsor secures project permissions.	Agreed consent matrix, legal opinions and environmental work.
Currency / energy / finance	SPV and shippers allocate/pass through or hedge defined exposures.	Stress tests, power arrangements and hedging policy.
Default / abandonment	Shareholder and financing documents specify step-in, termination and funded restoration.	Costed decommissioning and enforceable security.

No risk allocation is effective merely because it appears in a PPP term sheet. Confirm each counterparty's legal authority, financial capacity and ability to control the risk. An unbounded guarantee or optimistic demand forecast should not be used to conceal a weak project case.

09 / ENVIRONMENT AND DELIVERY

A sequence of decisions, not a launch date

Gate	Work and decision
0 / Mandate	Agree sponsor roles, capped feasibility funding, data ownership and independent advisors. Confirm no construction commitment.
1 / Demand and alternatives	Market soundings, route longlist, whole-system comparison, policy and regulator engagement. Reject the corridor if demand or alternatives undermine it.
2 / Survey and concept	Marine/land surveys, design basis, environmental baseline, community engagement, comparative landfalls and first defensible cost/schedule range.
3 / FEED and agreements	Front-end engineering design (FEED), material qualification, project-specific consent programme, EPC/operating strategy and conditional shipper contracts.
4 / Final investment decision	Independent assurance, consents, supply/demand contracts, fully funded financing and contingency, agreed liabilities and public approvals.
5 / Build and accept	Construction, verification, testing, emergency exercises and performance acceptance; operation begins only after required authorisations.

Environment and communities

Assess seabed habitats, fisheries, shipping/navigation, landfall ecology, cultural heritage, construction disturbance, marine access and cumulative impacts. Engage Caithness, Orkney, Shetland and Norwegian communities early. Identify benefits without claiming job counts or emissions savings before evidence exists.

Measure lifecycle emissions, hydrogen leakage and compressor electricity; compare with alternative uses of renewable power. Establish grievance and compensation processes, local procurement opportunities and a funded restoration plan. Existing Scottish guidance identifies environmental and wider regulatory interfaces [2].

Schedule discipline

Do not set an opening year before the critical path is established. Marine surveys, consents, technology qualification, customer conversion and financing can determine the sequence. Stage duration, annual spend and construction windows are outputs of the development programme, not assumed facts.

10 / DEVELOPMENT MANDATE

The next reviewable decision

Invite a limited feasibility partnership, with a right to stop, rather than a commitment to build.

Mandate item	Proposed requirement
Participants	Expressions of interest from an energy developer/operator, UK and Norwegian public counterparties, relevant Scottish/local bodies and independent technical advisers.
Budget control	A separately approved, capped development budget with work-package pricing, milestones, procurement rules and a funded exit plan. No monetary amount is set before scopes and bids.
Governance	Named accountable sponsor, independent assurance lead, conflicts register, reporting timetable and decision rights.
Core deliverables	Demand/open-season report; corridor alternatives; bathymetric/geotechnical scope; hydrogen design basis; environmental/consent plan; cost range; financial model; risk allocation.
Stop / redesign triggers	Insufficient bankable demand; competitive alternatives materially better; unacceptable environmental/safety constraints; unaffordable public exposure or unresolved rail dependency.
Success test	A credible, independently reviewed case for the next development stage. Passing feasibility does not authorise FID or provide a public funding entitlement.

Data room required before financing discussions mature

Maintain traceable source data, sponsor authority records, route/landfall maps, survey interpretations, qualification results, customer commitments, cash-flow and sensitivity models, consent correspondence, contracts, insurance strategy and lifecycle/decommissioning provisions. Separate verified evidence from scenario inputs and superseded designs.

Commercial intent

The 60/40 ownership concept and 30-year term are opening positions for negotiation. They should change if a competitively tested structure delivers better value, clearer accountability or lower public exposure. Participation discussions, approvals and actual capital subscriptions are future actions, not accomplished facts.

11 / SOURCES AND QUALIFICATIONS

Evidence register

Evidence, assumptions and status

[1] DESNZ. Hydrogen Infrastructure Strategic Planning: policy statement. 7 October 2025.
Open primary source | Accessed 5 November 2026.
Planning context and DESNZ decision role; does not establish this project's support eligibility.

[2] Scottish Government. Hydrogen: planning and consenting guidance - section 9. 5 February 2026.
Open primary source | Accessed 5 November 2026.
Pipeline safety and wider authorisation context. Project-specific applicability still requires advice.

[3] Norwegian Government. Feasibility study on a hydrogen value chain between Norway and Germany. 23 November 2023.
Open primary source | Accessed 5 November 2026.
Historical precedent for a conditional industry/government study, not proof of the proposed UK-Norway route.

[4] DNV. DNV-RP-F123 Hydrogen pipeline systems. Edition 2026-03.
Open primary source | Accessed 5 November 2026.
Public scope description only; full standard and referenced requirements must be obtained for engineering work.

How to read this prospectus

Citations [1]-[4] support limited policy, historical or technical context. Route concept, PPP ownership, operating term, commercial terms, delivery gates and all numerical sensitivities are independent proposals or calculations. None has been approved by a named authority, customer or lender.

The earlier rail and pipeline illustrations are conceptual artwork, not survey evidence or engineering drawings. This prospectus uses an authored corridor relationship diagram and does not treat an illustration as proof of separation, geology or feasibility.

Principal undefined quantities

Route length, initial capacity, pressure, diameter, terminal location, directional operation, capital and operating costs, emissions, tariff, financing terms, public commitment, counterparty identities and opening date remain undetermined.

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